ENGO 531/632
Advanced Photogrammetric and Ranging Techniques

Laboratory Assignment 2

Close-Range Bundle Adjustment Analysis Part 2

Date given: 2 October 2026

Date due: 8 October 2026 at 11.59
Late group submissions will not be accepted;
Late individual submissions lose 10% per day

Marks: This assignment constitutes 5% of your final grade.

Objectives
This lab builds on the previous assignment and has the following objectives:

1. Analyze the impact of intersection geometry on network quality.
2. Study and explain the effects of different datum definitions.
3. Diagnose and solve problems in a bundle adjustment.

For students in ENGO 531
This assignment is to be performed by the group formed for Lab 1.

For students in ENGO 632
This is an individual assignment.

Assignment
Part A
This part of the assignment will use the input data from Lab 1. Ensure that the image coordinate observation standard deviation is set to ±0.3 pixel.

1. In Lab 1, point 21 was identified as the most weakly determined point. Point 17 is also weakly determined. Explain the reason for the weakness.
2. Points 12 and 13 share something in common with the points mentioned above. However, their precision is much higher. Why?
3. Why is the precision of point 12 slightly better than that of point 13?
4. Points 9 and 20 are observed in three images, which improves coordinate precision. Why is the precision of point 20 slightly superior?
5. Which of these points (9 and 20) is superior in terms of gross error controllability? Explain.
6. The datum for this network has been defined with weighted control points. By how much has it been over-constrained?

Part B

This part of the assignment uses a new dataset that is available in D2L. It is a close-range network of targets observed with a pre-calibrated Samsung Galaxy S10 mobile phone camera. The units of image space, which is again left-handed, are pixels and the object space units are m. The image point coordinate standard deviation is ±0.7 pixel. There are no control points in this network, only unknown tie points. (Note that the CON file, even though empty, must be present for the software to work.) The datum is defined by parameter observations of the EOPs of the first image (image 4) and the distance (independently) measured between object points 25 and 28. The distance is 0.913 m, which is a common length for scale bars used in metrology. Use ±0.1 mm for the PC observation precision and ±0.0001° for the angle observation precision. The spatial distance standard deviation is ±1 mm. Make the CFG and DIS files, run the adjustment and answer the following questions:

7. What are n, u and r for this network?
8. Briefly comment on the overall network quality in terms of object point precision.
9. Perform a simulation in which the distance observation precision is now ±10 mm. What happens? Explain.
10. Reset the distance *a priori* standard deviation to ±1 mm and perform another simulation in which the EOPs of image 28 are constrained. (Image 4 will now be unconstrained.) What happens? What conclusion can be drawn from the two simulations?
11. The next three simulations are investigations into what can go wrong with a bundle adjustment. Reset the distance *a priori* standard deviation to ±1 mm and constrained EOPs to image 4.
    a. Use the new PHO file named `engo531_lab2_sim`. Two points in one image have been incorrectly labelled, which is a typical problem encountered in photogrammetry. Use the bundle adjustment results to identify the image in which they appear. What effects do these outliers have? Briefly explain how to you would handle such large outliers in a real dataset you collected without prior knowledge of the existence of errors?
    b. Use the original PHO file. Another common problem in bundle adjustments is poor approximate parameter values. Angles given in the incorrect quadrant is a frequent error. Change the ϕ angle for image 28 from 16° to -16°. What happens? What can be concluded from this simulation?
    c. Reset the ϕ angle for image 28 to its initial value of 16°. Change the κ angle for image 28 from -92° to 92°. What happens? What can be concluded from this simulation?

Note that as with Lab 1, your answers to the questions above should be supported with suitable tabular and/or graphical presentations of results.

Submissions are to be uploaded as a single PDF file to the D2L Dropbox by the specified date and time. For ENGO 531 students, only one group member needs to make a submission but be sure to indicate all members of the group on the title page.

END OF LAB 2